

DTSA 5501

Algorithms for Searching, Sorting, and Indexing

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 1 – CS Pathway	AI Engineer (16.7 / 100)	12.3 / 100
SPECIALIZATION Data Science Foundations: Data Structures and Algorithms Pathway		

12

CURRENT AVG SCORE
across 6 job roles

62

POTENTIAL SCORE
with recommended additions

+71

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course covers basics of algorithm design and analysis, as well as algorithms for sorting arrays, data structures such as priority queues, hash functions, and applications such as Bloom filters.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	16.7	61.0	+44.3	HIGH
Applied Scientist	14.9	78.4	+63.5	HIGH
Machine Learning Engineer	12.1	53.7	+41.6	HIGH
MLOps Engineer	11.0	53.2	+42.2	HIGH
ML Engineer	10.3	46.8	+36.5	HIGH
Data Scientist	9.1	79.8	+70.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Experiment Tracking 1 of 6 roles

tracking

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Experiment Tracking** — would broaden appeal to one additional role. Candidate terms: tracking.

3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

16.7 → 61.0 (+44.3 pts)

perception models learning software code systems based problems

Applied Scientist

14.9 → 78.4 (+63.5 pts)

discovery scientific computational research problems models learning systems

Machine Learning Engineer

12.1 → 53.7 (+41.6 pts)

sets learning analytics tools protection exploratory business mobile

MLOps Engineer

11.0 → 53.2 (+42.2 pts)

detection systems learning machine tracking tools models development

ML Engineer

10.3 → 46.8 (+36.5 pts)

discovery learning machine systems models world problems product

Data Scientist

9.1 → 79.8 (+70.7 pts)

toolkit sets product analytics scripting social problems

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.

